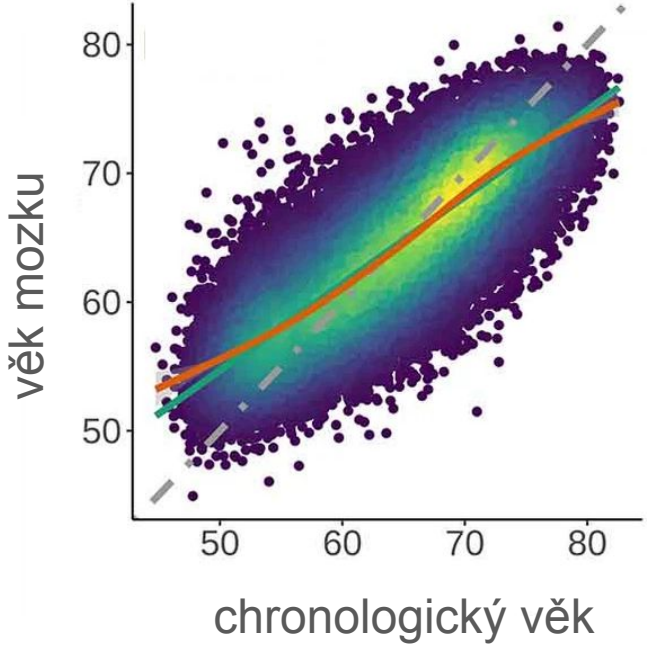
JAK STARÝ JE VÁŠ MOZEK?



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Ondřej Klempíř

DNAnexus

Ph.D. Biomedicínská a klinická technika

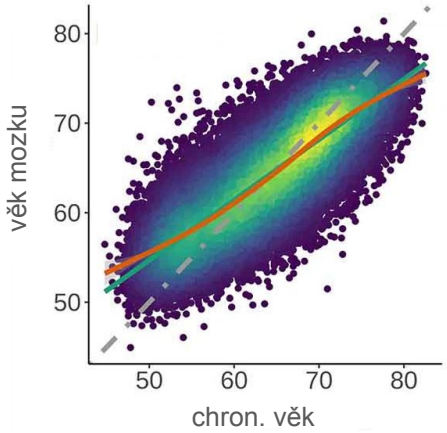
Analýza řeči
Neurozobrazování
Přesah do genomiky

Pharma, IBM, DNAnexus

Klempir, Ondrej

Czech Technical University in Prague, Prague, Czech Republic 57195508277 https://orcid.org/0000-0003-0773-5360

224	20	7	View all metrics >
Citations by ***210 documents***	Documents	*h*-index View *h*-graph	



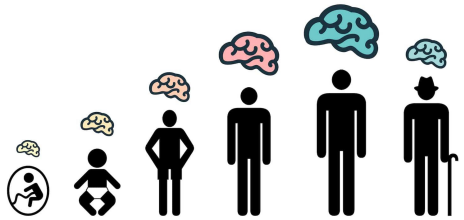
- *Jsme tak staří, jak se cítíme…?*
- *Brain-age* delta
- Zobrazovací techniky a algoritmy
- Biobanka
- Klasické metody regrese vs. machine learning
- Natrénujeme vlastní model

Studie o stárnutí a hodnocení progrese onemocnění

Stárnutí (ageing)

- Věk je hlavní faktor pro mnohá neurologická onemocnění
- brain-age delta vystupuje jako (bio)marker
- Sestavení modelu a jeho validace
- Rozdíly v pohlaví

Cumplido-Mayoral et al., 2023



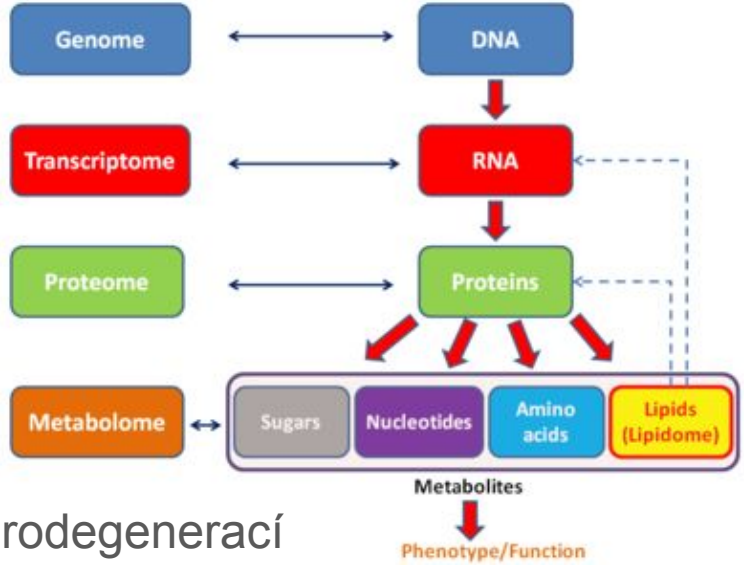
Parkinsonova nemoc

- Příznaky
  - Zamrzání
  - Poruchy řeči
  - Stuhlost
- Příklad studie
  - 2 kohorty (de-novo a chroničtí)
  - významně vyšší biologický věk vs. chronologický
  - cca. 3 roky rozdíl
  - Korelace s trváním nemoci, a rovněž i s kognitivní i motorickou zátěží

Eickhoff et al., 2021

Na základě čeho lze stanovit věk mozku?

- **Struktura (magnetická rezonance)**
- Funkce (funkční magnetická rezonance)
- Řeč
  - Řečové promluvy
- "Novinka": Proteomika
  - Kompozice proteomu krve a CSF*
  - Animální modely
  - Identifikace proteinů asociovaných s neurodegenerací



*CSF = mozkomíšní mok

Data z UK Biobank

- 0.5M+ subjektů
- Mozek, kardio, OCT …
- Mozková data opravdu rozsáhlá
  - 50K+ dosud
- Biobanky jsou typicky prostředek pro projekci menších dat a nálezů z laboratoří
  - Struktura biobanky se musí zohlednit

A note from UK Biobank's 50,000th scanned participant

YouTube

Browse by Primary Category of Origin

- Imaging
  - Abdominal MRI
    - Kidney MRI
    - Liver MRI
    - Pancreas MRI
    - Abdominal composition
    - Abdominal organ composition
  - Brain MRI
    - Scout images and configuration for brain MRI
    - T1 structural brain MRI
    - T2-weighted brain MRI
    - Arterial spin labelling brain MRI
    - Task functional brain MRI
    - Diffusion brain MRI
    - Resting functional brain MRI
    - Susceptibility weighted brain MRI
  - Heart MRI
  - DXA assessment

Field ID	Description
20227	Functional brain images - resting - NIFTI
20249	Functional brain images - task - NIFTI
20250	Multiband diffusion brain images - NIFTI
20251	Susceptibility weighted brain images - NIFTI
20252	T1 structural brain images - NIFTI
20253	T2 FLAIR structural brain images - NIFTI

4
2
1451

Brain Imaging Documentation

http://www.ukbiobank.ac.uk

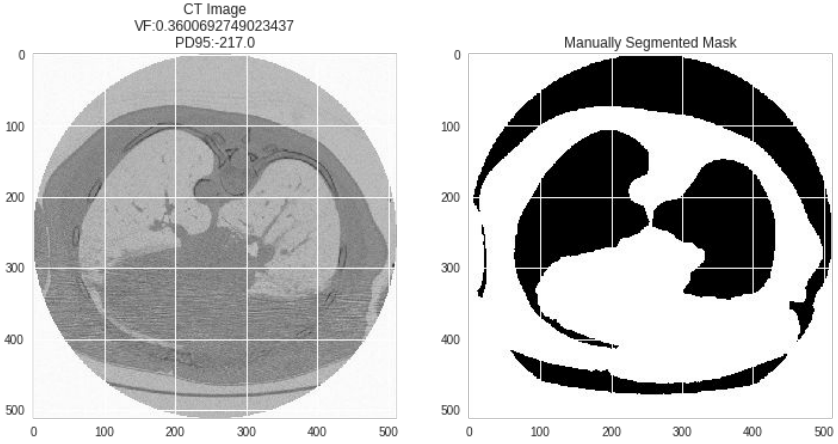
https://biobank.ctsu.ox.ac.uk/crystal/crystal/docs/brain_mri.pdf

biobank uk
Improving the health of future generations

https://www.dicomstandard.org

Pár slov o formátu DICOM

- Digital Imaging and Communications in Medicine
- Produkováno zobrazovacími systémy
- Data a metadata
- Vysoké rozlišení



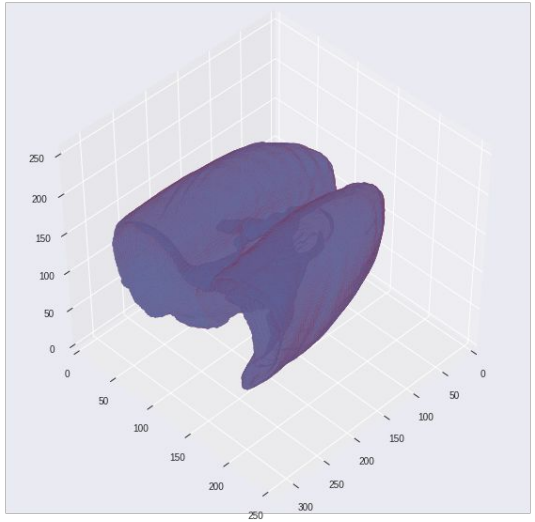
- 1D signály
- X-ray skeny
- CT
- MRI (T1 vs. T2 vs. FLAIR)

jeden soubor per řez (2D matice)

Pár slov o formátu NIFTI - voxely

- Neuroimaging Informatics Technology Initiative
- NIFTI je 3D struktura (tensor)
- Typicky jeden soubor obsahuje 3D informaci
- Kompaktní
- Preferováno některými ML nástroji

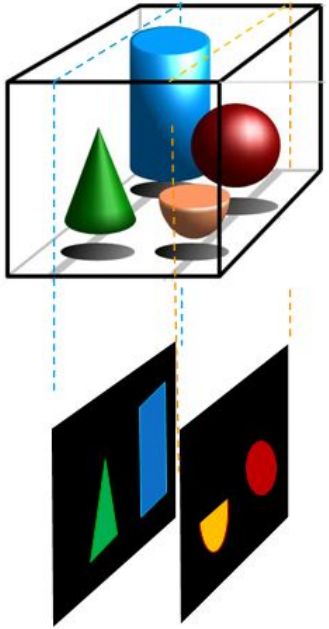
Příklad: MRI volumetrická data
Dimenze: (šířka, výška, index řezu)
Shape: (512, 512, 100)



Tomografie

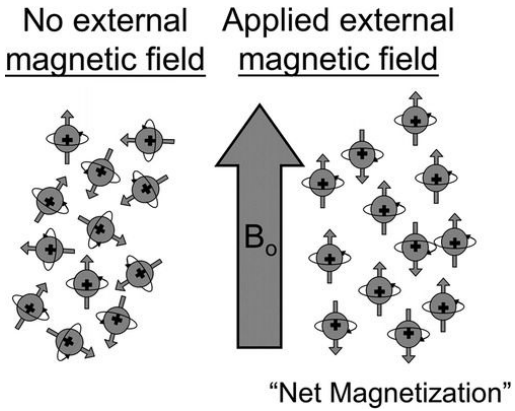
- Techniky pro zobrazení řezů skrze objekt (X-ray, ultrazvuk)
- Dokáže rekonstruovat 3D z 2D

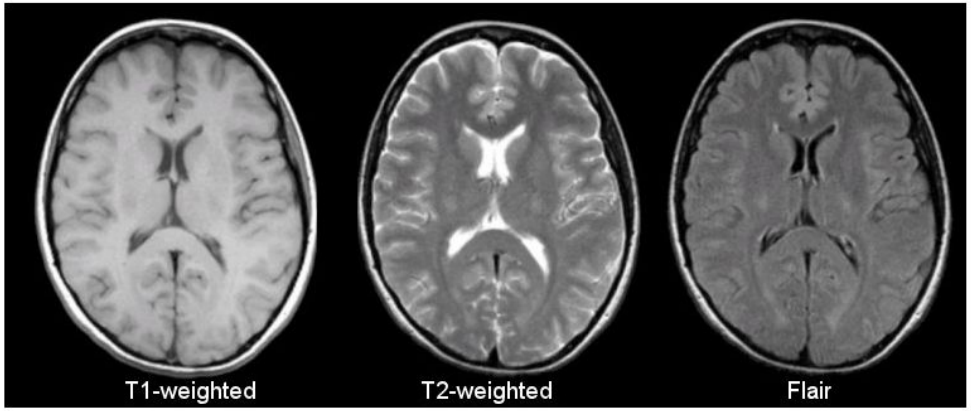
- **Počítačová tomografie (CT)**
  - Série obrazů z X-ray měření
  - CT skeny se ukládají jako 2D řezy (TIFF, JPEG)

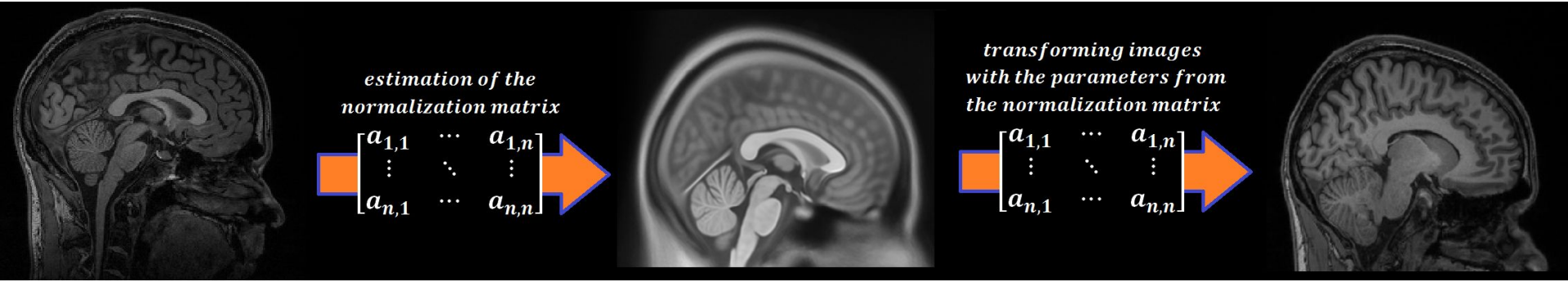


Zobrazení magnetickou rezonancí

- Výkonné magnety orientují jádra vodíku
- Kontrast a jas jsou určeny fyzikálními vlastnostmi tkáně
- T1 vs. T2 vs. FLAIR



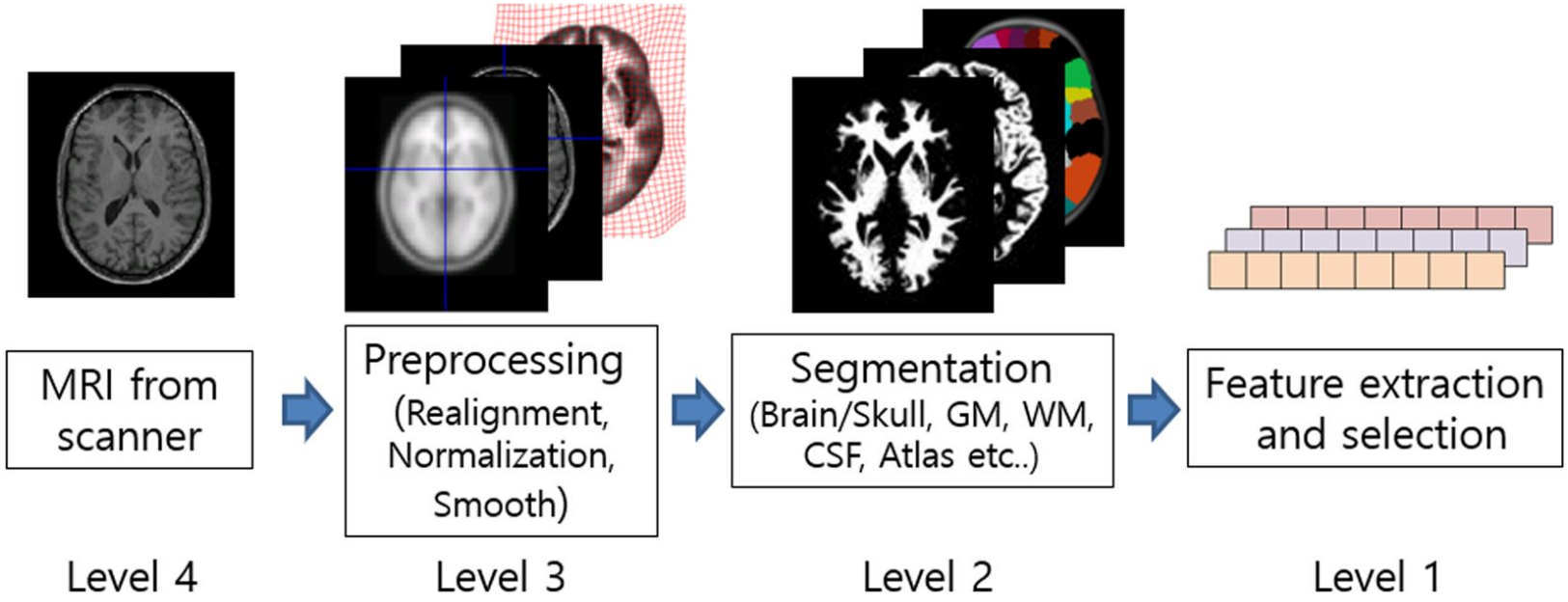




Jak zjistíte co vaše geniální metoda dělá na reálném vzorku?

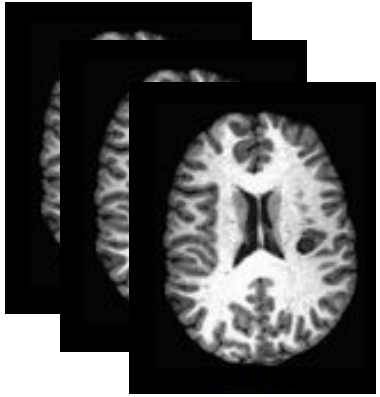
- ❑ Kontrola vizualizací

Pro klinické použití: manuální klasifikace fenotypů / anotace pro ML


MRI from scanner
Preprocessing (Realignment, Normalization, Smooth)
Segmentation (Brain/Skull, GM, WM, CSF, Atlas etc..)
Feature extraction and selection
Level 4
Level 3
Level 2
Level 1

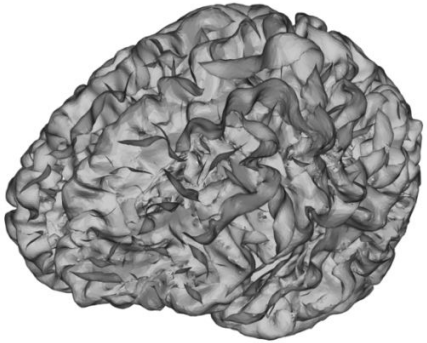
Oh et al., 2019

Co se děje v UKB - schematicky



DICOM

dicom2nifti



NIFTI

FSL

FreeSurfer

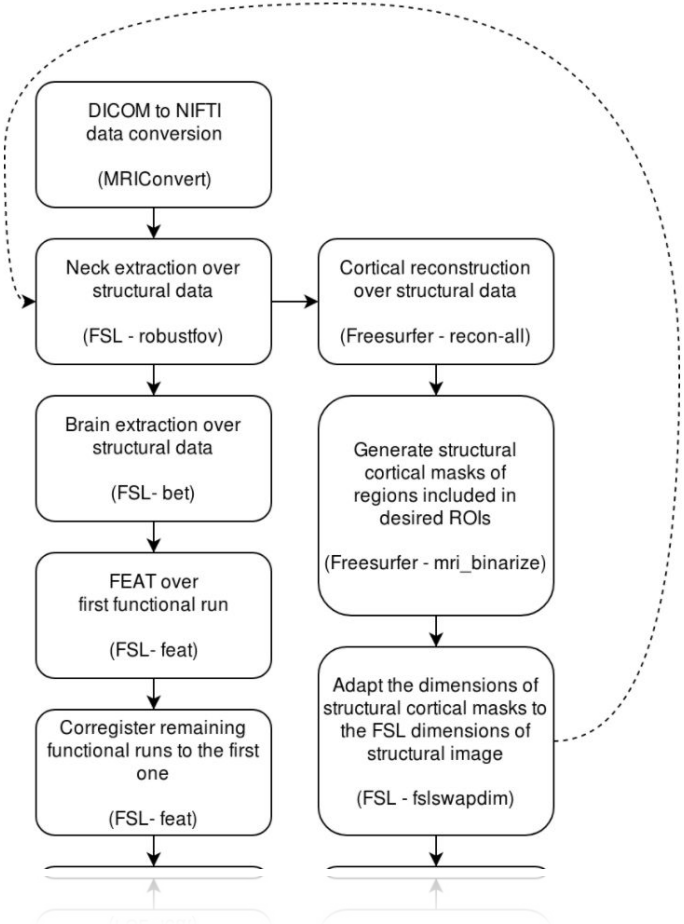
segmentace, registrace, ***IDPs***

Nářadí: co se hodí znát

- docker.io/brainlife/fsl:latest
- docker.io/nipype/nipype:latest

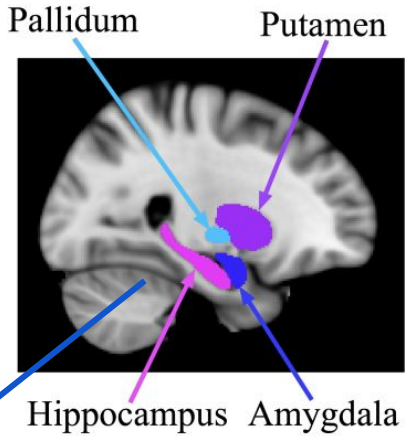
- pydicom - zpracování DICOM
- nibabel - zpracování NIFTI
- nilearn - statistika pro neuroimaging

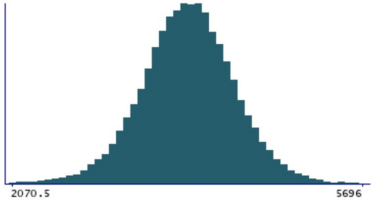
- https://git.fmrib.ox.ac.uk/falmagro/uk_biobank_pipeline_v_1.5



Parametry odvozené z obrazu

- Image Derived Phenotype (IDP)
- Je to číslo …
  - Skalární hodnoty odvozené z obrazu
  - Např. objemy jednotlivých oblastí ze strukturní MRI
- Vstup pro učící se systémy a analytiku
- Software vs. expertní přístup





Data-Field 25020

Description: Volume of hippocampus (right)

Category: Subcortical volumes (FIRST) □ T1 structural brain MRI □ Brain MRI □ Imaging □ Assessment centre

biobank uk

Index | Browse | Search | Catalogues | Downloads | Login | Help

Data-Field 25010

Description: Volume of brain, grey+white matter

Category: Assessment centre ▸ Imaging ▸ Brain MRI ▸ T1 structural brain MRI

Imaging measures + Brain MRI

Participants	46,535
Item count	51,182
Stability	Accruing

Value Type	Continuous, mm3
Item Type	Data
Strata	Derived

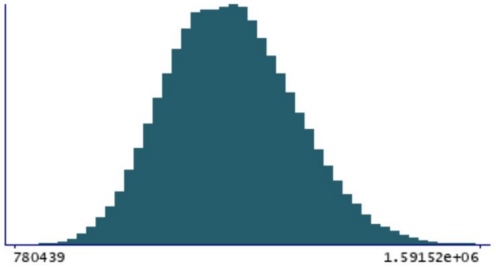
Sexed	Both sexes
Instances	Defined (2)
Array	No

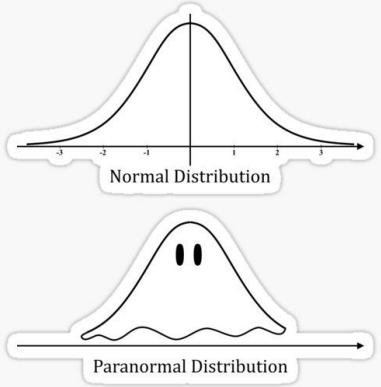
Debut	Oct 2015
Version	Feb 2023
Cost Tier	d1 o1 s1

Data | **2 Instances** | **Notes** | **0 Related Data-Fields** | **0 Resources**

51,182 items of data are available, covering 46,535 participants.
Defined-instances run from 2 to 3, labelled using Instancing 2.
Units of measurement are mm3.

Maximum	1.74371e+06
Decile 9	1.30439e+06
Decile 8	1.25119e+06
Decile 7	1.21341e+06
Decile 6	1.18156e+06
Median	1.1525e+06
Decile 4	1.12287e+06
Decile 3	1.09324e+06
Decile 2	1.06058e+06
Decile 1	1.01726e+06
Minimum	780439





Enabling scientific discoveries that improve human health

https://biobank.ndph.ox.ac.uk/showcase/field.cgi?id=25010

NATURE
An open access paper
www.nature.com

Nature. 2022; 604(7907): 697–707.
Published online 2022 Mar 7. doi: 10.1038/s41586-022-04569-5
PMCID: PMC9046077
PMID: 35255491

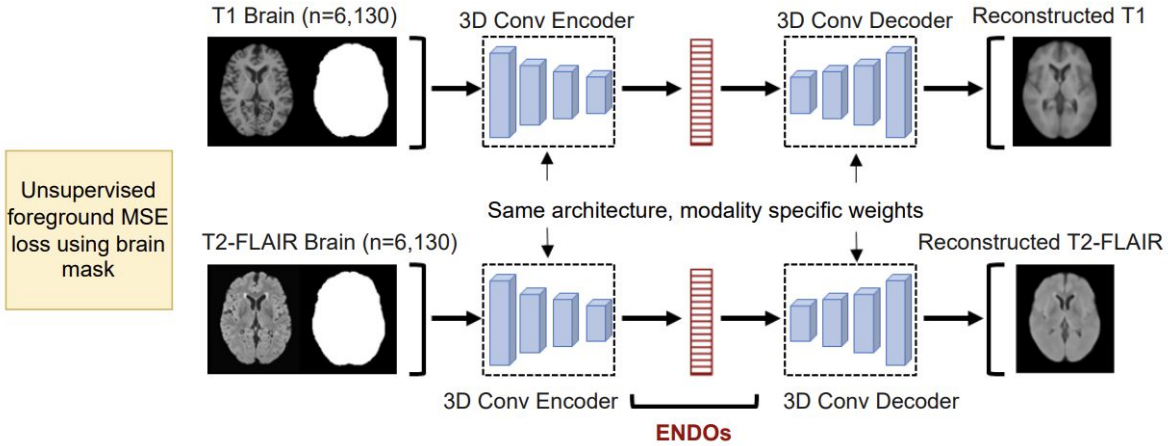
Journal Article

SARS-CoV-2 is associated with changes in brain structure in UK Biobank

Gwenaëlle Douaud,[1] Soojin Lee,[1] Fidel Alfaro-Almagro,[1] Christoph Arthofer,[1] Chaoyue Wang,[1] Paul McCarthy,[1] Frederik Lange,[1] Jesper L. R. Andersson,[1] Ludovica Griffanti,[1,2] Eugene Duff,[1,3] Saad Jbabdi,[1] Bernd Taschler,[1] Peter Keating,[4] Anderson M. Winkler,[5] Rory Collins,[6] Paul M. Matthews,[7] Naomi Allen,[6] Karla L. Miller,[1] Thomas E. Nichols,[8] and Stephen M. Smith[1]

První pokusy o stanovení parametrů strojovým učením

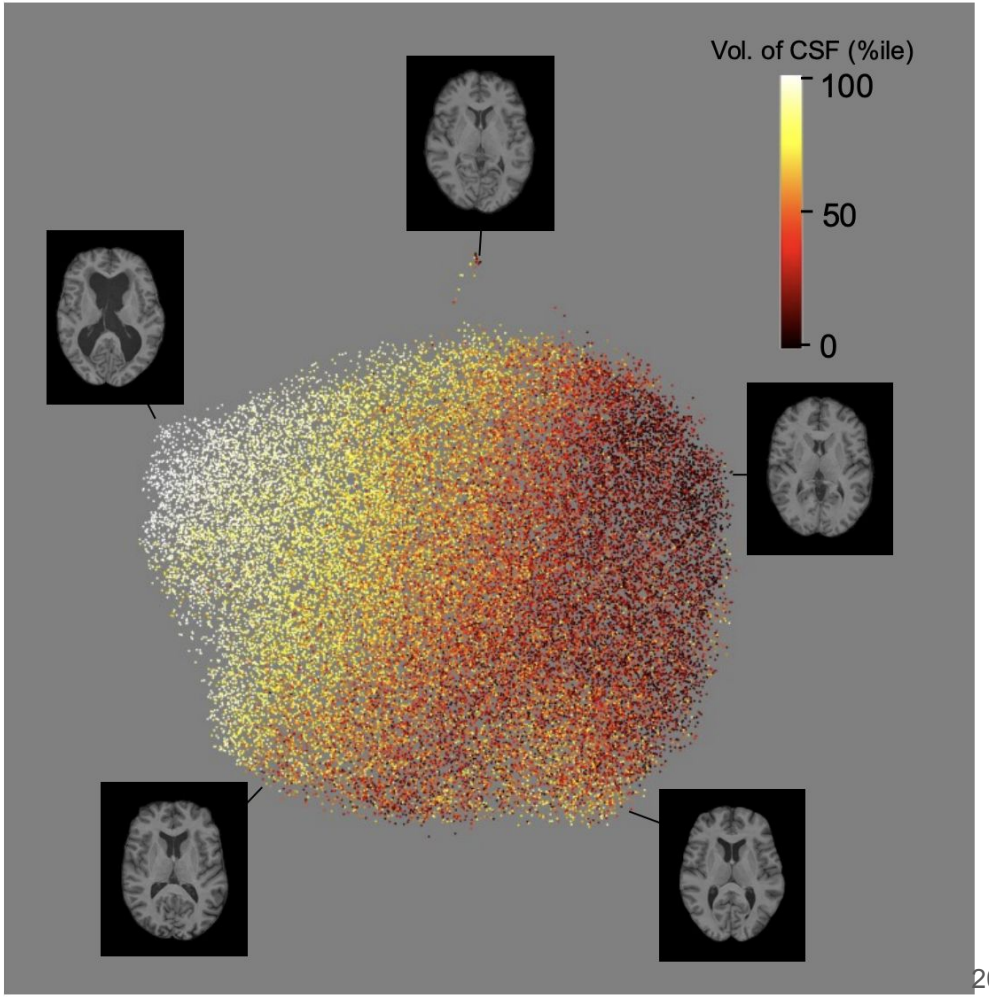
- Nesupervizované hluboké učení (učí se reprezentace)
- 3D konvoluční autoenkodér
- 128-dimenzionální reprezentace
  - “endophenotypy” (ENDOs)



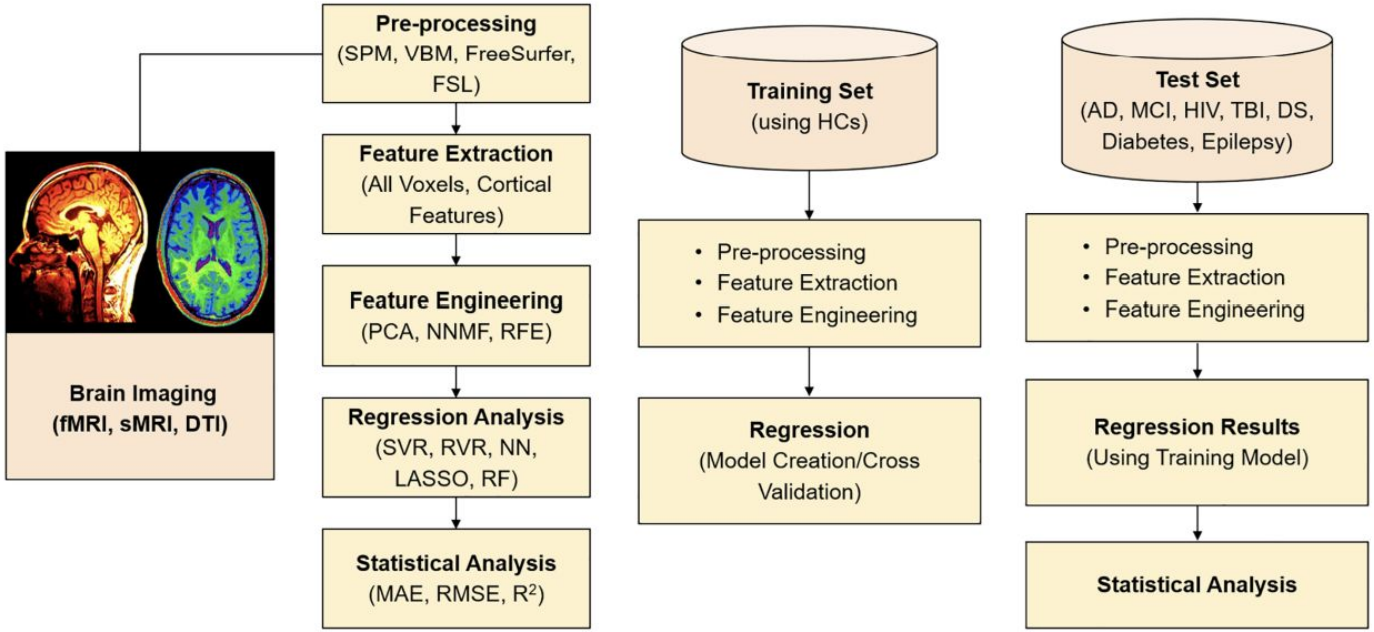
Patel et al., 2022

UMAP: redukce dimenze

Patel et al., 2022



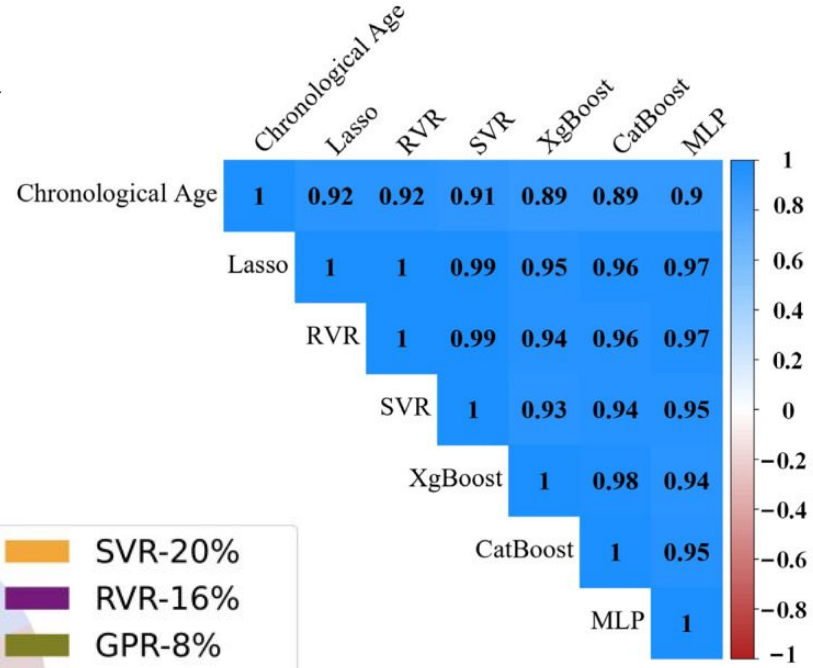
Metody pro “brain-age” delta modelování



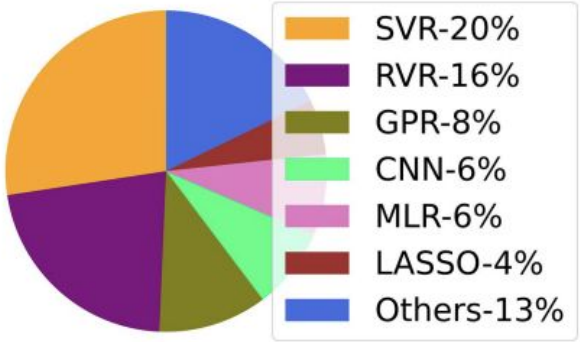
Mishra et al., 2021

Algoritmy strojového učení

- Lasso
- RVR
- SVR
- XgBoost
- CatBoost
- MLP

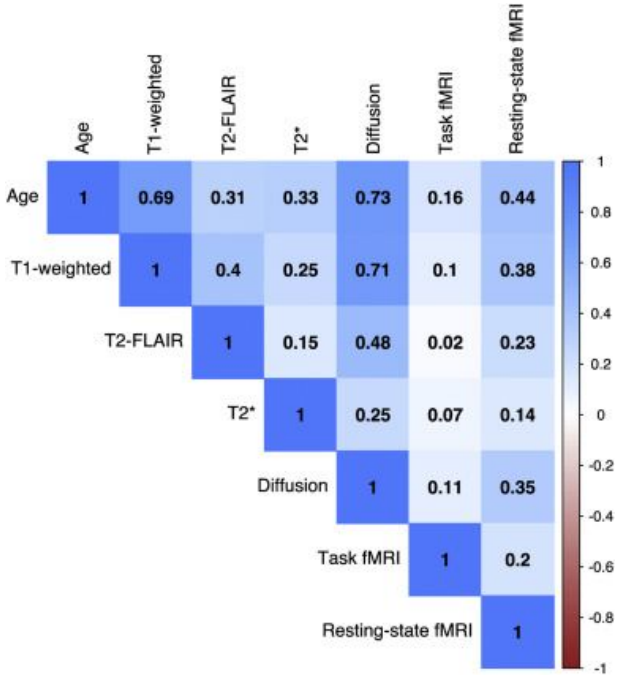


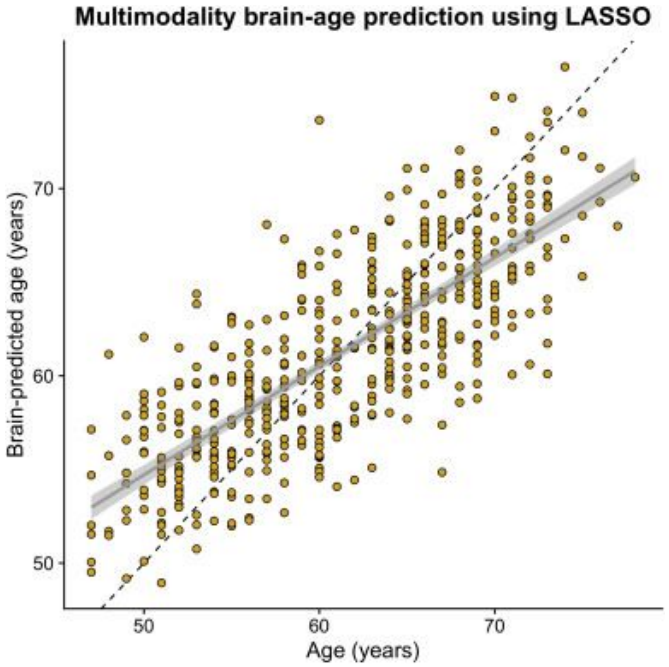
Xiong et al.,2023



Mishra et al., 2021

Jak přispívají jednotlivé modality





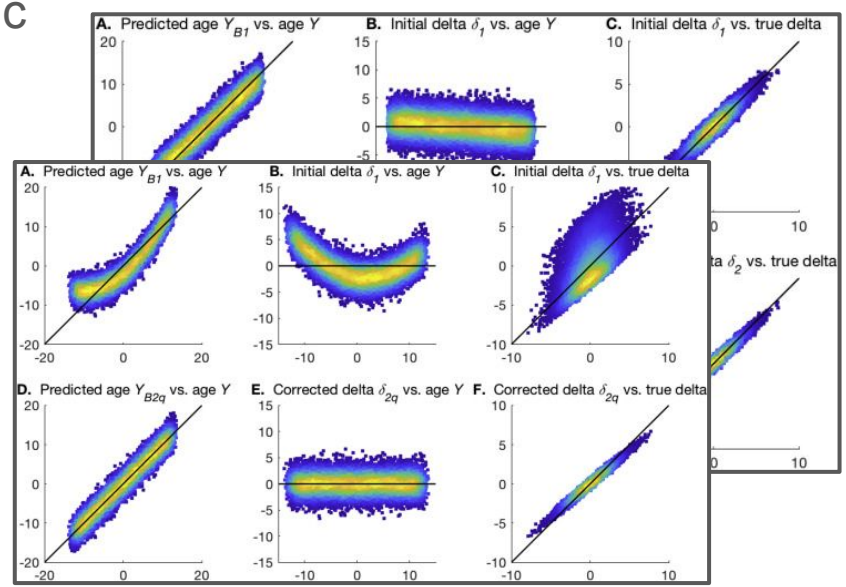
+ leave-one-modality-out scénář

Cole, 2020

Lineární vs. nelineární modely

Smith et al., 2019

- Článek přináší přehled základních rovnic
- Upozorňuje na základní problémy statistických metod a navrhuje korekce
- Používá simulovaná data a data z UK Biobank
- Ve výsledku v zásadě doporučují cross-validaci
- Rigorózně komentuje limitace



"One cannot assume that the effect of aging on imaging measures is a linear function of age; indeed, acceleration of the effects of aging (in older age) seems quite likely, particularly in disease."

JUNE 26, 2023 ONDREJ KLEMPIR, PH.D., SR. COMMUNITY ENGAGEMENT SCIENTIST, DNANEXUS & BEN BUSBY, PHD, PRINCIPAL SCIENTIST, DNANEXUS

Infrastructure Challenges: Machine Learning in the Cloud as a Future Oriented Solution Using UK Biobank

https://medium.com/dnanexus/infrastructure-challenges-machine-learning-in-the-cloud-as-a-future-oriented-solution-using-uk-6e10bedb995a

```
Total volume of white matter hyperintensities (from T1 and T2_FLAIR images) | Instance 3    0.38
Total volume of white matter hyperintensities (from T1 and T2_FLAIR images) | Instance 2    0.35
Volume of white matter | Instance 2                                                        -0.14
Volume of white matter | Instance 3                                                        -0.16
Volume of hippocampus (left) | Instance 2                                                  -0.23
Volume of hippocampus (right) | Instance 2                                                 -0.24
Volume of brain, grey+white matter | Instance 2                                            -0.26
Volume of brain, grey+white matter | Instance 3                                            -0.27
Volume of white matter (normalised for head size) | Instance 2                              -0.29
Volume of hippocampus (left) | Instance 3                                                  -0.29
Volume of hippocampus (right) | Instance 3                                                 -0.29
Volume of Hippocampus (left hemisphere) | Instance 2                                       -0.32
Volume of Hippocampus (right hemisphere) | Instance 2                                      -0.32
Volume of white matter (normalised for head size) | Instance 3                              -0.34
Volume of Hippocampus (left hemisphere) | Instance 3                                       -0.36
Volume of Hippocampus (right hemisphere) | Instance 3                                      -0.37
Volume of brain, grey+white matter (normalised for head size) | Instance 2                 -0.54
Volume of brain, grey+white matter (normalised for head size) | Instance 3                 -0.57
dtype: float64
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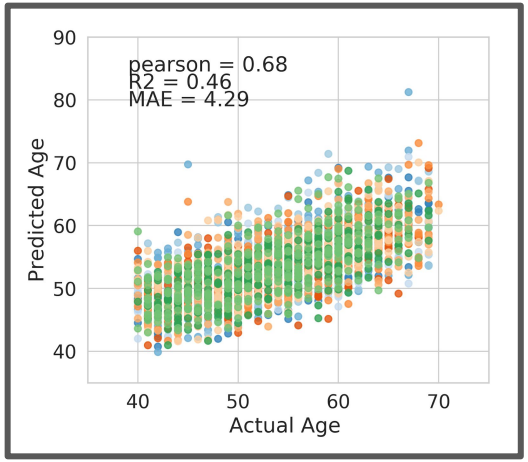

praktická část - příklad

DĚKUJI MOC, ŽE JSTE PŘIŠLI!

DNAnexus®

Materiály pro tuto přednášku vznikly v rámci UK Biobank Application **46926**.
Děkujeme všem účastníkům UK Biobank za poskytnutí dat!